

Pushpita Das Proma

Electrical & Electronic Engineering Student • Hardware & Software Developer

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PROFESSIONAL PROFILE

Motivated Electrical & Electronic Engineering student bridging hardware and software development. Experienced in microcontroller programming, embedded system prototypes, digital electronics logic, and modern web frameworks. Passionate about designing practical engineering solutions ranging from assistive tech to robotics and embedded display systems.

EDUCATION

B.Sc. in Electrical & Electronic Engineering

Jamalpur Science and Technology University (JSTU), Bangladesh

Present

Undergraduate

Higher Secondary Certificate (HSC)

Sherpur Govt College • Board: Mymensingh

2022

GPA: 5.00

Secondary School Certificate (SSC)

Sherpur Govt Girls' High School • Board: Dhaka

2020

GPA: 5.00

TECHNICAL SKILLS

Hardware & Microcontrollers:

ESP32, Arduino Uno, Sensor Interfacing, Gesture Control, Digital ICs (4511 Decoder), Servo Motors, LED Arrays

Engineering Domain:

Electrical Machines, Digital & Analog Electronics, Circuit Analysis, Embedded Systems, Power Electronics

Programming & Software:

C/C++, HTML, CSS, JavaScript, TypeScript, React, Git, GitHub

Tools & Simulation:

VS Code, MATLAB, COMSOL, PSpice, OrCAD, Arduino IDE

KEY PROJECTS

1. Patient Assistant System – Smart Gesture Gloves

Assistive Technology Project

Designed and built a smart gesture-controlled glove system using flex/motion sensors to translate hand gestures into actionable commands or auditory alerts for bedridden or speech-impaired patients.

2. Robotic Arm Using ESP32

Robotics & Embedded Systems

Engineered a multi-axis robotic arm controlled via ESP32 microcontroller, utilizing servo motors and wireless communication to perform precise picking, placing, and positional control.

3. BCD to 7-Segment Display Decoder Using IC 4511

Digital Electronics Hardware

Constructed a hardware digital logic circuit implementing the CMOS 4511 BCD-to-7-segment latch/decoder/driver IC to drive 7-segment LED displays from binary inputs.

4. 4x4x4 LED Cube Using Arduino Uno

Embedded Systems & Hardware Design

Designed and assembled a 3D 4x4x4 (64 LEDs) matrix cube driven by an Arduino Uno, utilizing multiplexing techniques and custom C/C++ code to display dynamic 3D light patterns and animations.